A Multi-Seed Evaluation of Augmentation and Loss Interventions in Fine-Tuning Real-Time YOLO Detectors: A Study on PASCAL VOC and KITTI

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Abstract

Augmentation and loss changes for fine-tuning real-time detectors are typically expected to gain a few tenths of a mAP point, the same order as seed-to-seed variation. We ask whether such small gains can be separated from noise under a typical three-seed fine-tuning budget and, if not, where the noise comes from. We evaluated seven training-time interventions and several training-free inference options, each changing one baseline setting. We used COCO-pretrained YOLO11n on PASCAL VOC 07+12 and YOLO11s and YOLO26n on KITTI. Decision rules were fixed in advance, and test sets were opened once. With three seeds, the design resolves only effects of $\gtrsim$1 point on VOC validation and 2.9–3.6 points on KITTI. On VOC, none of the seven interventions met the acceptance threshold (0/7). Post hoc, the drop under untuned Varifocal loss was traced mainly to negative-sample suppression; rescaling the classification loss gain recovered $\approx$82% of it. On KITTI, one of ten improvement tests passed. It reversed sign with last-epoch weights, and most of its gain came from one test scene group. We therefore read it as checkpoint-selection noise, not generalization. We quantify four noise sources: seed variance, validation sampling, single-seed screening and checkpoint selection. On 1000 validation images, the per-seed sampling error ($\approx$0.5 point) exceeds a 0.3-point screening threshold. Single-seed screening looks accurate even under null effects. Checkpoint selection shifted the mean intervention offset on YOLO11s by $\approx$4$\times$ the acceptance threshold. Each source comes with a practical recommendation. The results apply to untuned single-setting interventions against an untuned default recipe.

Keywords: object detection, YOLO, fine-tuning, reproducibility, seed variance, negative results

1 Introduction

Real-time object detection serves applications with latency budgets of a few milliseconds and devices with limited hardware. One of its main tools is the single-pass YOLO (You Only Look Once) family [1, 2]. In this family, the architecture can stay fixed while only the training recipe changes. A new augmentation, box loss or post-processing step is added, and the accuracy table shows a small increase. A single-run comparison cannot tell how much of such an increase is real. Part of it may come from the training seed and from model selection. In other areas of machine learning, improvement claims have often not been tested against a properly tuned baseline and a variance

estimate [3]. For PASCAL VOC, one study explicitly states that the official Ultralytics configuration selects the best checkpoint on the VOC07 test set [4]. Here, the best checkpoint (`best.pt`) means the weights saved during training according to the validation metric.

A common practical regime fine-tunes small pretrained weights on target data. Augmentation and loss settings do not touch the inference graph, so they look like the cheapest lever. Yet the expected gain and the noise have the same size. Consider the PASCAL VOC 07+12 fine-tuning of COCO-pretrained YOLO11n. The validation mean average precision (mAP; here mAP50-95) of the three-seed baseline has a standard deviation of 0.41 points. The differences expected from the interventions before the runs are only a few tenths. Our question is therefore: can small-gain claims be separated from noise under a typical fine-tuning budget? If not, where does the noise come from?

Proposed techniques were mostly evaluated at another model scale, on other data and with long training. Examples are masked Copy-Paste with large models [5] and YOLOX's advice to use weak augmentation for small models [6]. Their transfer to a short, box-labelled, nano-scale fine-tuning cannot be assumed. The reproducibility literature provides the necessary tools [7, 8]. In the sources we examined, however, no study applied these tools to real-time detector fine-tuning with multiple seeds, a locked test set and pre-specified decision rules (search scope: Section 2).

This study does not propose a new detector. It reports the protocol with pre-specified decision rules that we used to test small improvements, together with its results. Each intervention differs from the baseline in a single setting. It is judged with thresholds fixed before the runs. Interventions are selected on the validation set only, and the test set is used once after selection ends. We compare untuned single-point interventions with an untuned default recipe. Runs added after the locked test was opened, and deviations from the pre-specified rules, are disclosed separately (Sections 3.1 and 9). Our central thesis is that small-gain claims could not be separated from noise under a typical three-seed fine-tuning budget. We show this with seven interventions and two datasets. We also document four sources of noise: seed variance, validation sampling, single-seed screening and checkpoint selection.

On VOC, none of the seven training-time interventions met the acceptance threshold in three-seed validation (0/7). Box-based copy-paste (H03) was the only training intervention that entered the locked test. It led by +0.25 points on validation but trailed by −0.12 points on test. On KITTI, we ran ten improvement tests with YOLO11s and YOLO26n. Only one passed: multi-scale training (H01) on YOLO11s. This acceptance reverses sign with the last-epoch weights (`last.pt`). Most of its gain also comes from a single test scene group. We therefore read the KITTI experiment as an illustration of checkpoint-selection noise, not as evidence of generalization.

Our contributions are as follows:

1. We evaluate seven training-time interventions and several training-free inference options, together with their latencies. The protocol uses multiple seeds, a locked test set and pre-specified decision rules.
2. We document four sources of noise in claims based on small differences and derive a practical recommendation for each (Section 8).
3. We decompose the drop observed with untuned Varifocal loss into its components.

Appendix A summarizes the experiment log and gives calculation notes. Additional results and method details are in Online Resource 1, which is also available in the GitHub repository. The paper cites it as "Online Resource 1, Section X.n".

2 Related work

Real-time detection.

The YOLO family began with YOLOv1, which framed detection as a single-pass regression problem [1]. Later versions enriched the training recipe more than the architecture. For example, YOLOX introduced a decoupled head and an anchor-free design [6]. YOLO11 is an Ultralytics release without a peer-reviewed technical report [2]. The YOLO26 report introduces end-to-end inference without non-maximum suppression (NMS) and the MuSGD optimizer [9]. Our baseline inherits MuSGD as the package default. Our focus is not the architecture but the training and inference settings of small YOLO models.

Data augmentation and loss functions.

Mosaic combines four images into one sample. YOLOX recommends weakening strong augmentation for small models, including MixUp, which blends two images [6]. The idea of cutting and pasting objects goes back to Cut-Paste-Learn [10]. Ghiasi et al. show that, with segmentation masks, large models and long training on COCO, it is a strong augmentation [5]. Kisantal et al. paste small objects several times [11]. The built-in Ultralytics `copy_paste` augmentation is meant for segmentation and is off by default [12]. In box regression, DIoU and complete IoU (CIoU) add centre-distance and aspect-ratio terms [13]. Inner-IoU continues this line with auxiliary boxes [14]. CIoU is the default Ultralytics box loss [12], so gains of the "switch to CIoU" type are already in our baseline. VarifocalNet proposes the Varifocal Loss (VFL), which scales the class target by IoU [15]. GFL proposes quality-focused classification and the distribution focal loss (DFL) for boxes [16]. TOOD's task-aligned assigner (TAL) uses the class score and IoU together [17]. PP-YOLOE uses VFL with TAL in its standard recipe, so this pair is not expected to be harmful by itself [18].

Inference-time operations and small objects.

Soft-NMS lowers the scores of overlapping boxes instead of deleting them [19]. DIoU-NMS adds the centre distance to the suppression criterion [13]. Weighted boxes fusion (WBF) was designed to merge boxes from different models [20], so its use with a single model needs its own test. Test-time augmentation (TTA) also changes the inference resolution. Its gain is therefore hard to separate from the augmentation itself [21]. Most of these methods were not evaluated under a real-time latency constraint. We test them for a single model under a pre-specified relative latency budget. For the small-object gap, small-object-focused augmentation has been proposed [11].

Reproducibility, seed variance and the gap.

Lipton and Steinhardt list trends that hide the true source of an improvement [3]. Picard shows that seed variance is modest but that outlier seeds are easy to find [8]. Bouthillier et al. propose a comparison protocol that accounts for sources of variance [7]. Cawley and Talbot show that overfitting the selection criterion can inflate performance estimates [22]. We borrow the idea of fixing decisions before seeing results from preregistration practice [23]. We found no official document on which data the v8.4.165 defaults were tuned; that they were tuned on COCO is an assumption. YOLO-PEFT states that its results under the official VOC configuration are not single-opening blind test estimates [4]. VOC results that follow this configuration may therefore be optimistic. We could not check this for the other sources we use as anchors (Section 4.1).

In the sources we examined, no study in real-time detection tested an intervention against a validated baseline with multiple seeds, a locked test set (single opening) and pre-specified decision rules. This is not a claim that no such study exists; it reflects the scope of our search. The search was not systematic. We found and verified sources through Crossref, arXiv, the Semantic Scholar API, publisher pages and web search. The main search ran from 2026-09-29 to 2026-10-02. Three sources on the KITTI experiment and later analyses were added between 2026-10-03 and 2026-10-22. The search queries were not recorded.

3 Method

In the fine-tuning regime, the gain expected from an intervention is as large as the variation due to the training seed. Under this condition, deciding after looking at the results can cause selection bias [7, 22]. The protocol therefore fixes the decision before the runs [23]. The hypothesis and thresholds are written down, the baseline is validated, runs are staged and the test set is opened once (Fig. 1). All rules but one were written before the runs. The exception is the KEEP rule, which was frozen in round 2, after the H01–H03 results (Section 3.3). There is no public, time-stamped registration; details are in the Limitations (Section 9).

3.1 Analysis framework with pre-specified decision rules

Accuracy values are in points (mAP×100). $M_{c,s}$ is the validation mAP50-95 of configuration c trained with seed s, with IoU thresholds from 0.50 to 0.95 in steps of 0.05. B denotes the baseline, k an intervention card and $S = \{0, 1, 2\}$ the seed set, so $n = |S| = 3$. The seed-paired difference and its

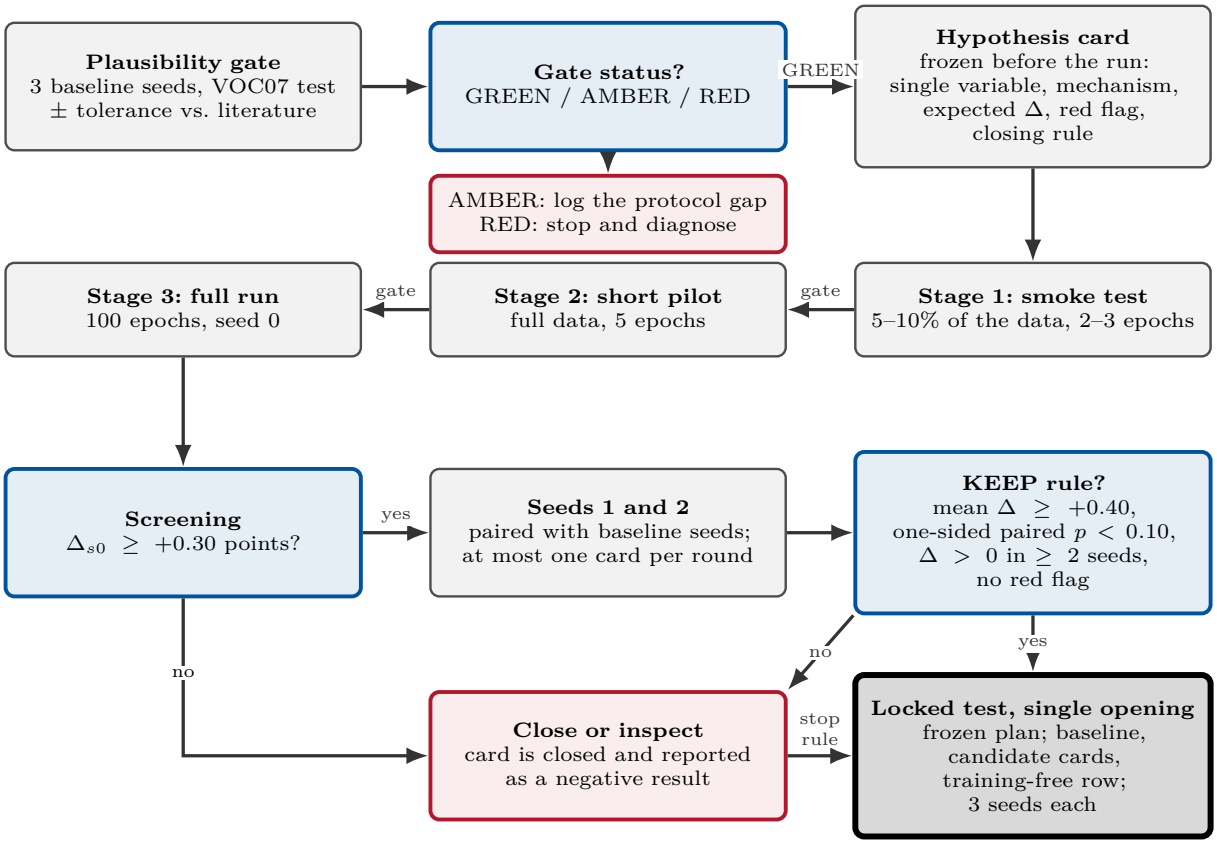


Figure 1 Flow of the analysis protocol with pre-specified decision rules. Blue-framed boxes are decision gates (written before the run), red boxes are stopping or card closure, and the thick frame is the locked test.

mean are

$$\Delta_{k,s} = M_{k,s} - M_{B,s}, \qquad \bar{\Delta}_k = \frac{1}{n} \sum_{s \in S} \Delta_{k,s}. \quad (1)$$

Hypothesis card.

Each intervention is defined by a hypothesis card written before the runs. The card fixes the single variable, the mechanism, and the expected direction and size of the effect. It also fixes a red-flag threshold, above which a result counts as "implausibly good", together with the closing rule, the budget and the secondary metrics. The upper part of a card is never changed afterwards; corrections require a new card. In practice, this rule offers limited evidence (Section 9; full card rule: Online Resource 1, Section D.1). If the red flag is exceeded, the card is not accepted, and we audit leakage, the split and the baseline. There are three decision labels. KEEP means the card is accepted, CLOSE means it is closed, and INSPECT means it needs further inspection.[1] Table 1 lists the variable and the pre-specified expectation of the eight cards. The ranges come from the literature or from the closest anchor.

Plausibility gate and staged runs.

We compare the VOC07 test mAP50-95 and mAP50 of the three-seed baseline with a range published under a similar protocol (Section 4.1). A GREEN result moves on to the intervention stage. An AMBER result documents the protocol difference. A RED result stops the runs. The tolerance is never widened afterwards. The gate tests only the baseline. The baseline therefore sees the test set once, and its values appear in the gate report and the experiment log. We disclose this. Implementation errors are caught in cheap stages. Stage 1 is a smoke test (5–10% of the data, 2–3 epochs).

[1] The original labels in the study's records are the Turkish TUT, ÖLDÜR and İNCELE. They are rendered here as KEEP, CLOSE and INSPECT without change of meaning.

Table 1 The eight hypothesis cards. Expected Δ and red flag are pre-specified values of the validation mAP50-95 difference [points]. For H07, the values vary by row family.

Card	Family	The only change from the baseline	Expected Δ	Red flag
H01	Scale	batch resolution range m: $0 \to 0.25$	$[0, +0.5]$	$> +1.0$
H02	Box loss	CIoU $\to$ Inner-CIoU, $r = 0.75$	$[0, +0.5]$	$> +1.0$
H03	Copy-paste	box-based, $p = 0.5$, $\sqrt{A} \leq 48$ px	$[0, +0.4]$	$> +0.8$
H04	Scale	in-image scale amplitude σ_s: $0.5 \to 0.25$	$[0, +0.5]$	$> +1.0$
H05	Class loss	BCE $\to$ Varifocal, $\alpha = 0.75$, $\gamma = 2$	$[-0.5, +0.5]$	$> +1.0$
H06	Box loss	box loss gain λ_{box}: $7.5 \to 10$	$[-0.3, +0.3]$	$> +0.8$
H07	Training-free inference	NMS threshold, Soft-NMS, test resolution, TTA, WBF (12 rows)	by family $[0, +1.0]$	$> +1.0 \ldots +2.0$
H08	Copy-paste dose	$p = 1.0$, $\sqrt{A} \leq 96$ px	$[0, +0.8]$	$> +1.0$

Single-variable exceptions: H08 changes the probability and the candidate pool together (Section 3.2). On YOLO26n, H02 and H05 are applied to both training heads (Section 7.1).

Stage 2 is a short pilot (full data, 5 epochs). Stage 3 is full training (100 epochs). After each stage, we check an expectation file written before the runs, with metric bounds, a variant signature and counter ranges. On a violation, the queue stops. In stage 3, a live safety rule stops a run if a NaN appears or if the validation metric falls below a lower bound at a given epoch. Ultralytics disables mosaic from the start when `epochs` ≤ `close_mosaic`. The stage 1–2 runs of H03 and H08 therefore use `close_mosaic=0` and are compared with a baseline that uses the same setting.

Test lock.

Interventions are selected on the validation set only. During the loop, the metric files contain no test key. For the interventions, the test set was opened only in the locked step at the end of the loop (K5), on 2026-10-01 at 21:23:45. The baseline values, in contrast, were measured and seen on 2026-09-30 for the plausibility gate. The first measurement ran at 10:16:22; the anchor file had been written on 2026-09-29 at 22:21:13. The gate report (16:10) and the experiment log record these values. The first rows of seed 0 were seen earlier than planned, at 10:40, in an analysis step before the gate report (Section 9). The same weights gave the same values in K5 (Section 4.1).

According to the records, three things happened between the two measurements. The H04–H08 cards were written, the KEEP rule was frozen, and the safety threshold of H05 was recalibrated. The writing notes state that these steps used validation information only. The range and tolerance of the plausibility gate did not change. In our view, the baseline test values carry no information about the effects of the interventions; they only give the absolute level of the baseline. For the H04–H08 cards and the KEEP rule, written after the first measurement, the risk is limited to this information. The K5 plan (runs, comparisons, statistics) was written before the test runs. After that, no new hypothesis, training or hyperparameter search was run on the test set.

The loop moves to K5 under any of three conditions. A card receives KEEP; the Δ of the best training card stays below +0.30 in two consecutive rounds; or the budget or the round limit is reached. The budget was 75 planned graphics processing unit (GPU) hours with a 90-hour cap, and the round limit was 5. The K5 runs need no training. The baseline, the card closest to the threshold on validation (H03) and the training-free TTA row enter with three seeds each (nine runs). The other cards are evaluated on validation only. After unlocking, we did some additional work on VOC, all on validation only. It covers seed 1–2 runs of the six screened-out cards, the WBF diagnostic grid, power analyses and the H05 mechanism cards. None of it was applied to the test set. The CLOSE labels were given before the additional seeds and have not changed (full list: Appendix A.2).

Implementation checks.

We searched for code errors with several checks:

- identity unit tests for the loss changes, i.e., an exact match with the baseline loss for Inner-CIoU at $r = 1$ and, on negative cells, for VFL at $\alpha = 1$, $\gamma = 0$;
- a double-precision gradient check of the VFL positive branch against a hand-written implementation;
- WBF unit tests (8/8);
- a head counter on YOLO26n that confirms the patches reach both training heads;

- the expectation files of the staged runs and automatic health gates.

In revisions, a number-integrity script checked that every number in the text also appeared in the previous version. The authors checked all code, analyses and text produced by AI agents.

3.2 Baseline and interventions

Baseline.

The baseline is Ultralytics YOLO11n [2]. The library is pinned to version 8.4.165 (commit `004db34`), and its code is untouched [12]. The network has 2.59 million parameters and 6.52 GFLOPs. Its head is anchor-free and decouples classification from box regression. Training starts from the COCO-pretrained `yolo11n.pt` weights [24]. The training loss is a weighted sum of three terms:

$$\mathcal{L} = \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{dfl}}\mathcal{L}_{\text{dfl}},$$
$$(\lambda_{\text{box}}, \lambda_{\text{cls}}, \lambda_{\text{dfl}}) = (7.5,\ 0.5,\ 1.5). \tag{2}$$

The TAL assigner selects the positive anchors [17]. For anchor i and ground-truth (GT) box j, the alignment metric is $a_{ij} = s_{ij}^{0.5}\, u_{ij}^{6}$. Here s_{ij} is the score of the GT class and u_{ij} the CIoU between prediction and GT. For each GT, the 10 anchors with the highest metric become positive. The class target of a positive, $t_{i,c} \in [0, 1]$, is the metric rescaled to the highest IoU of its GT. The target thus carries both class confidence and localization quality. Let $\mathcal{P}$ be the set of positive anchors, $\hat{b}_i$ the predicted box, b_i the assigned GT box and $z_{i,c}$ the class logit. With $w_i = \sum_c t_{i,c}$ and $T = \sum_{i,c} t_{i,c}$,

$$\mathcal{L}_{\text{box}} = \frac{1}{T}\sum_{i\in\mathcal{P}} w_i\big(1 - \text{CIoU}(\hat{b}_i, b_i)\big),$$
$$\mathcal{L}_{\text{cls}} = \frac{1}{T}\sum_{i}\sum_{c} \text{BCE}(z_{i,c}, t_{i,c}), \tag{3}$$

where binary cross-entropy (BCE) is applied to each class independently. The CIoU [13] and DFL [16] terms are written out in Online Resource 1, Section D.2. Data augmentation follows the Ultralytics default. It uses mosaic with probability 1.0, RandomPerspective with scale amplitude $\sigma_s = 0.5$, HSV shifts and horizontal flips. MixUp and the built-in copy-paste are off, and mosaic is switched off in the last 10 epochs (details: Online Resource 1, Section D.3). The built-in copy-paste needs masks, so it has no effect on box-labelled VOC [12]. H03 and H08 therefore use a new box-based implementation. In all cards below, the loss, the assigner and the augmentation pipeline stay unchanged apart from the tested variable.

Scale axis (H01, H04).

H01 enables batch-level multi-scale training [25]. With $m = 0.25$, each batch is rescaled to a square size $s_b \in \{480, 512, \ldots, 800\}$, chosen with equal probability among multiples of the stride ($g = 32$). Validation stays at 640 pixels. H04 narrows the in-image scale amplitude ($\sigma_s : 0.5 \to 0.25$). The rationale is the advice to weaken augmentation for small models [6]. The direction follows a fallback rule written before H01 was run.

Box loss (H02, H06).

H02 replaces the CIoU term in Eq. (3) with Inner-CIoU [14]. The edges of the predicted and GT boxes are shrunk by the ratio r around their centres. With IoU_i^{in} the IoU between these auxiliary boxes,

$$\mathcal{L}_{\text{box}}^{\text{inner}} = \frac{1}{T}\sum_{i\in\mathcal{P}} w_i\Big[\big(1 - \text{CIoU}_i\big) + \text{IoU}_i - \text{IoU}_i^{\text{in}}\Big],$$
$$r = 0.75. \tag{4}$$

At $r = 1$, the loss equals that of the baseline. At $r < 1$, the gradient steepens for high-IoU samples, so the card targets precision at high IoU thresholds. H06 stays in the same family but tests the weight instead of the form: $\lambda_{\text{box}} = 7.5 \to 10$.

Classification loss (H05).

H05 replaces the BCE term with Varifocal-weighted BCE [15]. With $y_{i,c} = \mathbf{1}[t_{i,c} > 0]$,

$$\mathcal{L}_{\text{cls}}^{\text{VF}} = \frac{1}{T}\sum_{i}\sum_{c} w_{i,c}^{\text{VF}}\,\text{BCE}(z_{i,c}, t_{i,c}),$$
$$w_{i,c}^{\text{VF}} = \alpha\,\sigma(z_{i,c})^{\gamma}\,(1 - y_{i,c}) + t_{i,c}\,y_{i,c}, \tag{5}$$

with $\alpha = 0.75$ and $\gamma = 2$. On negatives, the weight is $\alpha\sigma(z)^{\gamma}$, so the gradient of easy background samples is suppressed. On positives, the weight is the target quality t. The normalization $(1/T)$ and $\lambda_{\text{cls}} = 0.5$ are the same as in the baseline.

This choice reduces the total classification gradient. Under the single-variable rule, α, γ and λ_{cls} are not tuned. H05c is a separate, single-variable mechanism card (Section 6).

Box-based copy-paste (H03, H08).

After mosaic, the method acts with probability p. It selects one to three candidate objects on the mosaic canvas with $\sqrt{A_j} \leq \tau_A$ that do not overlap other boxes. Each object is pasted without rescaling, by alpha blending, at a free location that does not intersect existing boxes. A new box of the same size is added [10, 11]. H03 uses $p = 0.5$ and $\tau_A = 48$ pixels; H08 uses $p = 1.0$ and $\tau_A = 96$ pixels. H08 thus changes the probability and the pool together. The dose D (pasted objects per mosaic) is measured from counters. For H08, a target of $D \geq 0.80$ was pre-specified. A dose below the target is not tuned, to avoid adding a second variable (steps and dose gate: Online Resource 1, Sections B.2 and D.4).

Training-free inference family (H07).

The weights are fixed (baseline seed 0 `best.pt`). The reference pipeline R0 uses a confidence threshold of 0.001, an NMS IoU threshold of 0.7, at most 300 detections and a single scale of 640 pixels. Each row differs from R0 in one component only, and rows are not combined. The components are the NMS threshold (N), Soft-NMS (S) [19], the test resolution (U; 704, 768 or 832 pixels), TTA [12, 21] and WBF (W) [20]. T1 is the Ultralytics default TTA, and T2 uses upscaling scales (definitions: Online Resource 1, Section D.4). For each family, a candidate row is selected on one half of the validation set. It counts as CONFIRMED if it meets three pre-specified conditions: on the other half, in an image bootstrap and on the baseline's seed 1–2 weights (Online Resource 1, Section B.3). The latency ratio is $\rho = L_{\text{row}}/L_{R0}$. A CONFIRMED row counts as GAIN-A (within the real-time constraint) if $\rho \leq 1.10$, and as GAIN-B otherwise. GAIN rows do not receive KEEP. The training-time cards do not change the inference graph, so their latency equals that of the baseline.

3.3 Decision rules and statistics

Screening and KEEP.

The seed-0 run is not evidence. It only earns the right to run the other seeds, which requires $\Delta_{k,0} \geq +0.30$ points. A card below this is closed. This means "no detectable improvement at $n = 1$", not "no effect". A card that passes screening is run with seeds 1 and 2. At most one card per round moves on to this stage. KEEP requires all of the following: (i) $\bar{\Delta}_k \geq +0.40$; (ii) one-sided $p < 0.10$ for the paired $t = \bar{\Delta}_k \sqrt{n}/s_\Delta$ (2 degrees of freedom); (iii) $\Delta_{k,s} > 0$ in at least two seeds; (iv) the red flag is not exceeded. CLOSE applies if $\bar{\Delta}_k < +0.15$ or if $\Delta_{k,s} \leq 0$ in at least two seeds. Cases in between are INSPECT. KEEP is a candidate count, not evidence. The paired t is reported uncorrected, and the final check is on the locked test set. The H01–H03 cards were written before the KEEP rule and carry a different in-card rule. With three seeds, $\bar{\Delta}_k < +0.30$ gives CLOSE. If $\bar{\Delta}_k \geq +0.30$ but $\Delta_{k,s} \leq 0$ in at least two seeds, the result is INSPECT. This rule has no t condition. Because the cards were not changed, the round-2 rule was not applied retroactively. H03 does not receive KEEP under either rule. H07 falls outside these rules (details: Online Resource 1, Section D.1).

Statistics.

Each difference is taken against the baseline of the same seed. For TTA the pairing is exact, because the weights are the same. For training cards the pairing is nominal. Even with the same seed, an intervention consumes the random number generator (RNG) of the augmentation differently, so pairing reduces variance only partly. On the locked test, the paired t-test is two-sided. The pre-specified main claim is a single comparison, H03 versus the baseline; TTA and the others are exploratory. We apply the Holm correction [26] to the twelve-test K5 family (Holm-12). We also apply it, as an exploratory step, to the validation differences of the seven cards (Holm-7). On test, we give an image-level paired bootstrap interval [27]. For training cards, we give a seed-level t interval (Online Resource 1, Section D.5). We use the word "significant" only for a Holm-corrected $p < 0.05$. On test, the 95% confidence interval (CI) must also exclude zero. An uncorrected $p < 0.05$ is called "uncorrected" and marked with † in the tables. Differences in mAP50, mAP75, $\text{AP}_S/\text{AP}_M/\text{AP}_L$ and at class level are exploratory.

Resolving power.

We express resolving power as a critical effect: $\delta_{50} = t_{0.975,2}\, s_\Delta/\sqrt{n}$. This is the effect detectable with 50% power, i.e., the half-width of the 95% t interval. On the locked test, H03 has $s_\Delta \approx 0.21$ and $\delta_{50} = 0.51$ points. On validation, the s.d. of the seed differences is 0.12–0.79, so the per-card δ_{50} is 0.31–1.97 points. With the pooled $s_\Delta = \sqrt{\frac{1}{7}\sum_k s^2_{\Delta,k}} \approx 0.49$, we get $\delta_{50} \approx 1.2$ (95% uncertainty interval 0.88–1.90). The critical effect at 80% power, computed from the noncentral t distribution, is 1.58 points. The three-seed design therefore reliably resolves only effects of $\gtrsim$1 point on VOC validation. Smaller effects cannot be ruled out. Nonparametric paired tests cannot reach a decision in this design. At $n = 3$, the smallest one-sided p of a sign-flip test is $1/2^3 = 0.125$, which is above the 0.10 threshold. Decisions therefore rely on the normality assumption of the paired t test. On KITTI, the decision rule includes only seed variance. A hierarchical bootstrap that resamples seeds and test scene groups together is post-unlock and exploratory (Section 7.2).

Level of evidence.

Table 2 groups the main results into three categories by level of evidence. The codes come from a claim audit.

4 Setup and baseline plausibility

Data.

Experiments use the PASCAL VOC 2007+2012 detection data [28]. The training set has 15 551 images and 37 651 objects. The validation set has 1000 images (2407 objects). They were drawn from the trainval pool with `random.Random(0)`, stratified by VOC07/VOC12. Selection relies on this set only. The test set is the official VOC2007 test set (4952 images, 12 032 objects). The locked test stage (K5) ran once, following the pre-specified plan (Section 3.1). `difficult` objects are removed from the labels (15.1% in training, 17.0% in validation, 19.7% in test).

Training and hardware.

We chose YOLO11n because it allows many three-seed runs on a single GPU with 8 GB of VRAM and belongs to the real-time nano class. Training uses 640 px, 100 epochs, batch 16, 4 data workers, seeds 0/1/2 and `deterministic=True`. All hyperparameters are Ultralytics defaults. With more than 10 000 iterations (here $\lceil 15\,551/64 \rceil \times 100 = 24\,300$; nominal batch 64), the `auto` optimizer selects MuSGD with a learning rate of 0.01 and ignores `lr0` [12]. MuSGD is the hybrid Muon–SGD optimizer of the YOLO26 report, with scales 0.5/0.5 [9]. "Default" means the default of library version 8.4.165. We did not verify that it matches the original YOLO11 recipe. All runs used one machine: Intel i5-12400F, 15.9 GB RAM, NVIDIA RTX 5050 (8 GB, driver 610.47), Windows 10, Python 3.11.0 and torch 2.14.0+cu130 (CUDA 13.0). One VOC training run takes ≈2.9–3.4 GPU hours. The VOC experiment (eight cards, three rounds, locked test, post-test extra seeds) took 60 runs and 75.45 GPU hours (breakdown and KITTI times: Appendix A).

Metrics and speed.

AP is computed in the 101-point COCO format [24]. We use a confidence threshold of 0.001, NMS IoU of 0.7 and at most 300 detections per image. mAP50-95 averages over IoU thresholds 0.50:0.05:0.95; mAP50 and mAP75 use a single threshold. AP_S, AP_M and AP_L are computed by box area ($< 32^2$, 32^2–96^2, $> 96^2$ pixels). VOC has no official size classes, so these values are informative only. Validation contains 143 small objects, and the seed range of the baseline AP_S is 2.87 points. Latency is measured end to end with batch 1, FP16 and PyTorch eager. It covers pre-processing, the model or TTA, NMS/Soft-NMS/WBF and the copy to CPU. We use 50 warm-up and 300 measurement images from the validation list and take the median of three interleaved rounds. R0 runs at 8.45 ms in the H07 validation session. The baseline runs at 9.43 ± 0.25 ms (106 frames per second, FPS) in the K5 test session. The constraint is a relative latency budget (ratio ≤ 1.10), not an absolute real-time criterion such as 30 FPS.

4.1 Baseline plausibility check

The baseline was measured once on the VOC07 test set with three seeds. It then passed through a gate with a tolerance around a literature anchor. Table 3 gives mAP50-95 as 63.49 ± 0.11 and mAP50 as 83.51 ± 0.17 points. The anchor range is 61.0–63.5

Table 2 Level of evidence of the main results. PS: decision based on a pre-specified rule. PS-secondary: pre-planned analysis that produces no decision. E: post-unlock or exploratory analysis.

Category	Results (and deviation, if any)	Code
(1) Pre-specified, followed as written	VOC: closing of H01, H02, H04 and H06 at screening; closing of H03 under the round-1 rule and its locked-test result; baseline plausibility gate; results of the within-budget H07 rows; locked-test comparison of TTA (planned, exploratory secondary).	PS; TTA: PS-secondary
(2) Pre-specified, with deviation	Closing of H05 (re-run instead of a new card after a false alarm); closing of H08 (run after the stopping rule); move to the locked test (the Δ definition in the stopping rule was ambiguous; TTA was kept outside the rule); decisions from round 2 on (the KEEP rule was frozen after H01–H03); KITTI S1 decisions, `last.pt` robustness label, clustered bootstrap and three-class subset (the rule document changed before any card Δ was seen; H07 latency was corrected); the H05-pos/H05-neg answer to S2 (run under a second rule document after the VOC lock was opened).	PS; KITTI secondary: PS-secondary
(3) Post-unlock or exploratory	Seed 1–2 runs of the six screened-out cards and the three-seed differences based on them (including the Holm-7 result of H05); screening probabilities; power analyses; WBF diagnosis; H05c; the validation–test mAP50 gap of H03; on KITTI, the hierarchical bootstrap, scene-group and tram sensitivity, and the common checkpoint shift.	E

Untested explanations (e.g., possible causes for H05, checkpoint explanations on KITTI) are not in this table. The text calls them "hypotheses". Deviations are given in category (2).

Table 3 Baseline plausibility check: VOC07 test mAP50-95 and mAP50 [points]. The literature rows were not produced under the same conditions. Their sources are cited separately. No color coding is applied (reference values).

Row	mAP50-95 [points]	mAP50 [points]
This work, YOLO11n (COCO-pretrained, 100 ep), seed 0	63.45	83.54
This work, YOLO11n (COCO-pretrained, 100 ep), seed 1	63.61	83.67
This work, YOLO11n (COCO-pretrained, 100 ep), seed 2	63.41	83.33
This work, mean ± s.d. (n=3)	63.49 ± 0.11	83.51 ± 0.17
Literature: YOLO11n from scratch, 500 ep [29]	61.3	81.5
Literature: YOLO11-N, 600 ep (general setting) [30]	61.0	81.2
Literature: YOLOv8n from scratch, 500 ep [29]	60.2	80.9
Literature: YOLOv8n, 200 ep† [31]	63.5	83.6
Literature: YOLO11s full fine-tuning (checkpoint selected on test) [4]	64.28	83.44
Acceptance range (literature range ± tolerance)†	59.8–64.7	80.0–84.8
Plausibility gate decision	GREEN	

† The upper anchor 63.5 (upper end of acceptance 64.7) belongs to YOLOv8n. It probably comes from a checkpoint selected on VOC07 test. The value was verified in the text of Section 4.4 of the source, because the table cell was unreadable. The other literature rows also differ in initialization, epochs and checkpoint selection. They serve as context only. Our values are 101-point COCO AP, with `difficult` objects excluded. They are not directly comparable to VOC07 11-point devkit values [28].

and 81.2–83.6. The acceptance range widens it by a tolerance of ±1.2 points, and the gate returned GREEN. The anchor set, the tolerance and the gate commands were specified approximately 12 hours prior to the first test measurement. In the locked test stage, the same three weights were measured again and gave the same values (63.45, 63.61, 63.41). The evaluation pipeline is deterministic.

The lower end of the anchor comes from YOLO11n trained for 500–600 epochs (61.0 and 61.3) [29, 30]. The upper end is a YOLOv8n value (63.5) trained for 200 epochs, whose checkpoint was probably selected on the test set [31]. The tolerance is twice the seed standard deviation reported by YOLO-PEFT (at most 0.6 points) [4] (Appendix A.6). The GREEN decision therefore rests on this upper anchor, which has a different architecture and is probably test-selected. Our value is 2.2 points above the highest YOLO11n anchor (61.3). With YOLO11n anchors only, the upper bound of the acceptance range would be $61.3 + 1.2 = 62.5$. Our value lies above it, so the gate would have returned AMBER. The acceptance range (59.8–64.7) is wide, so the gate is a loose check. It is also not an exact replication.

Our training set is smaller by the 1000 validation images, our checkpoint is selected on validation, `difficult` objects are removed, and we fine-tune for 100 epochs from COCO-pretrained weights (protocol differences of the anchors: Appendix A.6). Part of the gap is therefore expected, but the effect was not measured. We read the GREEN result as confirming that the baseline is technically sound and that subsequent negative results do not stem from an underperforming baseline.

5 VOC results

5.1 Validation and test

Table 4 gives the seven training cards and the training-free TTA row as seed-paired Δ. The baseline scores 63.07 ± 0.41 points on validation and 63.49 ± 0.11 on test. None of the seven interventions met the KEEP criterion in three-seed validation (0/7). The card with the highest mean is H03 (+0.25 ± 0.24; one-sided $p = 0.107$). It did not beat the baseline on the locked test either (−0.12 ± 0.21; $p = 1.000$ after Holm-12). An uncorrected $p < 0.05$ appears only for H02 and H05. After Holm-7 (exploratory) only H05 remains, and its direction is negative (−2.87 ± 0.33). The absolute values of the cards and their seed 1–2 means are in Online Resource 1, Table B1.

In Fig. 2a, the means of five cards stay within twice the baseline s.d. band (0.81 points). In H04, seed 1 is an outlier (−1.76). H05 lies between −2.53 and −3.19 in all three seeds. All seven cards were closed with CLOSE. The intervention was active in all three seeds of every card, so the near-zero results do not come from an unconnected hook. No card's 95% t interval lies inside the ±0.30 margin, and for five cards the interval covers both +0.30 and +0.40. We therefore do not say "ineffective". We say "$\Delta \geq +0.30$ could not be shown in this regime". The latency ratios of the training-time cards are 0.99–1.04 (we make no speed claim). For no card does ΔAP_S exceed the seed range of the baseline. In the error-type breakdown, background false positives are the costliest type in both sets, and the error profiles of H03 and H06 did not shift. When the dose rose from 0.142 (H03) to 0.671 (H08), together with the candidate pool, the validation gain did not grow. In the qualitative panel, the effect of TTA goes both ways from image to image (Online Resource 1, Sections B.9–B.11).

5.2 Training-free inference and TTA

Fig. 3 plots the Δ mAP50-95 of the H07 rows against the latency ratio (row values: Online Resource 1, Table B3). None of the six training-free rows inside the band (NMS IoU, Soft-NMS, 704 px) beat R0. Their differences range from −0.58 to −0.10. Under this constraint, training-free post-processing gave no gain.

Test resolution gives an opposite-direction trade-off. As the input grows from 704 to 832 px, AP_S rises (U704 +2.41, U768 +4.55, U832 +4.94) and AP_L falls (−0.82, −3.41, −7.24). mAP50-95 drops from −0.31 to −5.19. The model was trained at 640 px, so the loss on large objects is consistent with a train–test scale mismatch (not tested). This is only an order of magnitude (single weights, 143 small objects, no CI). The small-object gap appears related to resolution, and the training-time interventions did not close it. A scale-aware inference that re-runs only images with small-object candidates at high resolution might limit the AP_L loss (not tested).

TTA is the only positive row. On validation (seed 0) it gives +0.80 (×2.57; 8.45 → 21.68 ms). On test it gives +0.83 ± 0.22 (×2.29; 9.43 → 21.64 ms, 46 FPS). At 46 FPS it is still real-time in absolute terms; what it exceeds is the relative budget. Its test p is 0.022 uncorrected and 0.222 after Holm-12, so it is not significant. The image-bootstrap CI ([+0.50, +1.14]) does not include training-seed variance. The cost is −1.07 in AP_M and −1.37 in precision at the operating point. The gain concentrates at high IoU and on large objects (mAP75 +1.12, AP_L +1.50) and is absent in mAP50 (+0.10). The smallest Holm-adjusted p among the twelve tests is 0.112, so these secondary differences are read as directions only (Online Resource 1, Table B2). We therefore read TTA as an exploratory accuracy–latency trade-off, not as an improvement. Adding the default WBF setting to the T1 branches changed mAP50-95 by −0.28 relative to R0 (−1.08 relative to T1) and raised mAP50 by +1.03. WBF was designed for model ensembles. For the scale branches of a single model, the fusion setting must be chosen separately (not tested on test; exploratory diagnosis: Online Resource 1, Section B.7).

Table 4 VOC: seed-paired Δ mAP50-95 vs. baseline [points], n=3. Baseline: validation 63.07 ± 0.41, test 63.49 ± 0.11. Color: Holm-significant only (blue: better, red: worse).

(a) Validation					
Card	Change	Δ±s.d.	p	95% t interval	Screening (s0)
H01	multi-scale 0.25	−0.21±0.61	0.614	[−1.72, +1.30]	+0.07 → CLOSE
H02	CIoU → Inner-CIoU	−0.35±0.12	0.041†	[−0.66, −0.04]	−0.22 → CLOSE
H03	box copy-paste	+0.25±0.24	0.214	[−0.35, +0.85]	+0.37 → passed; CLOSE
H04	scale amplitude 0.25	−0.85±0.79	0.204	[−2.82, +1.12]	−0.30 → CLOSE
H05	BCE → Varifocal	*−2.87±0.33*	0.004	[−3.68, −2.05]	−2.88 → CLOSE
H06	box loss gain 10	+0.04±0.56	0.912	[−1.35, +1.43]	−0.42 → CLOSE
H08	copy-paste, dose	−0.08±0.40	0.754	[−1.07, +0.90]	+0.22 → CLOSE
TTA	TTA (training-free)	+0.81±0.41	0.075	–	–

(b) Test (locked VOC07 test; only comparisons that entered the test)					
Card	Comparison	Δ±s.d.	p	Holm-12 p	95% CI
H03	box copy-paste	−0.12±0.21	0.416	1.000	[−0.40, +0.16]
TTA	TTA (training-free)	+0.83±0.22	0.022†	0.222	[+0.50, +1.14]

p: two-sided paired t-test. Acceptance: $\bar{\Delta} \geq +0.40$ and one-sided $p < 0.10$. Screening: s0 $\Delta \geq +0.30$. Color: Holm-7 on validation; on test, Holm-12 and a CI that excludes zero (image bootstrap). †: uncorrected $p < 0.05$. The raw decision labels "TUT"/"ÖLDÜR"/"İNCELE" appear here as KEEP/CLOSE/INSPECT.

5.3 Validation–test consistency

The seed-level t interval of H03 on test ([−0.63, +0.39]) covers the +0.30 margin. An improvement of +0.30 points therefore cannot be ruled out. The validation–test gap is 0.37 ± 0.28 (paired $p = 0.146$), so the sign change is consistent with sampling noise. H03's mAP50 gain was positive in all three seeds on validation (+0.74 ± 0.16, $p = 0.015$). On test it is −0.02 ± 0.21 (gap 0.76; $p = 0.036$, uncorrected, exploratory). The validation-source explanation (VOC07/VOC12) was not supported, and `best.pt` selection optimism was not tested. The residual remains unexplained (Online Resource 1, Section B).

6 Mechanism of the Varifocal loss (H05)

The mechanism notes for the other cards are brief (details: Online Resource 1, Section B.1). On the scale axis (H01, H04), the intervention was active, but mAP50-95 did not rise. H04 was +0.56 ahead of the baseline in the 5-epoch pilot but fell to −0.85 at 100 epochs. For the box loss (H02, H06), the targeted AP_{75} rose neither with Inner-CIoU (−0.42 ± 0.27) nor with the box loss gain (+0.07 ± 0.30). The weight of DFL, the largest loss share, was not tried. For copy-paste (H03, H08), the mAP50 gain fell from +0.74 ± 0.16 to +0.25 ± 0.14 when dose and pool were widened together. Because the two variables were not separated, the "saturated mechanism" reading is a weak hypothesis.

Symptom.

H05's mAP50-95 loss is the only difference that remains after Holm-7 ($p = 0.031$). It also lies outside the pre-specified expectation range (−0.5 to +0.5). Since PP-YOLOE uses VFL and TAL together in its recipe [18], this is a violation of expectation, so we first audited the implementation and the mechanism. The loss appears in all three size classes and is larger in mAP50 (−4.44). This suggests that the loss lies on the detection and classification side rather than in localization precision. The mechanism is active: the VFL gradient norm is 0.34 of the BCE norm (0.342, 0.341 and 0.341 in the three seeds; Appendix A.5). Because λ_{cls} stayed fixed, the effective classification signal weakened. The implementation passed two unit tests (Section 3.1). The result concerns untuned VFL. It does not generalize to "VFL is harmful" or "TAL and VFL are incompatible".

First-run failure.

The first full H05 run was stopped at epoch 5 by the live safety rule (exit code 95; validation mAP50-95 below the 0.2538 threshold). The threshold had

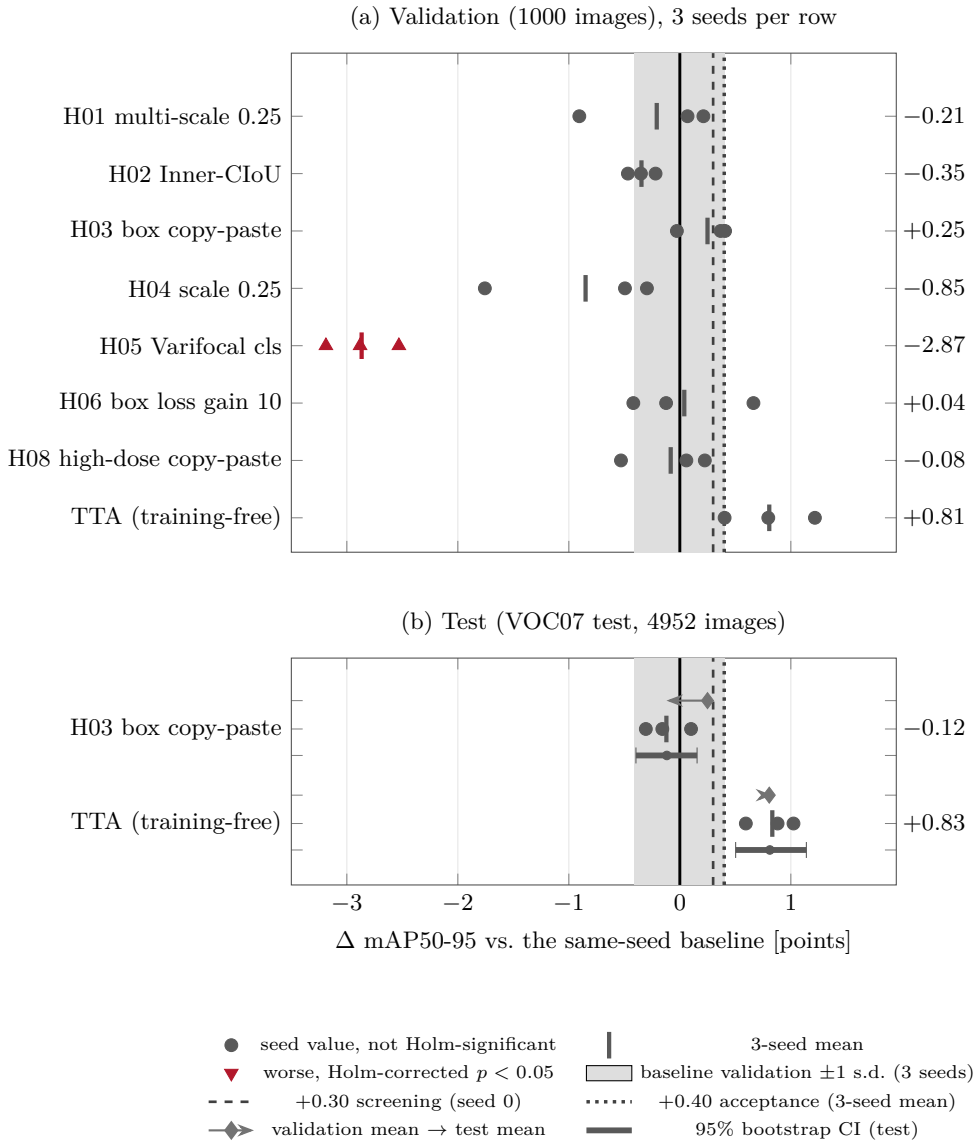


Figure 2 Per-card Δ mAP50-95 vs. baseline [points], three seeds: (a) validation, (b) test. The test CI covers image resampling only. Shading: ±1 baseline seed s.d. Lines: +0.30 screening and +0.40 acceptance thresholds. Red triangle: Holm-significant worse; gray: not significant.

been inherited from the BCE curve of H02. VFL dips deeper during warm-up, and the curve was rising, so the alarm was false. The card was re-run once with a rule recalibrated for VFL before the run (validation < 0.40 at epoch 20). The reported values, including seeds 1 and 2, come from this configuration. Re-running without opening a new card is a deviation (Section 9). The practical lesson is that live safety thresholds should not be inherited from another loss family. They should be calibrated in advance from the variant's convergence profile in the short pilot.

Possible causes.

The seed-paired validation difference is −7.87 points at epoch 20, −4.43 at 50, −2.81 at 90 and −2.63 at 100. Epoch 100 refers to the last epoch, whereas Δ refers to the best checkpoint. The difference stays negative in all three seeds but narrows. A narrowing gap does not look like an implementation bug with a fixed offset, but it is not decisive.

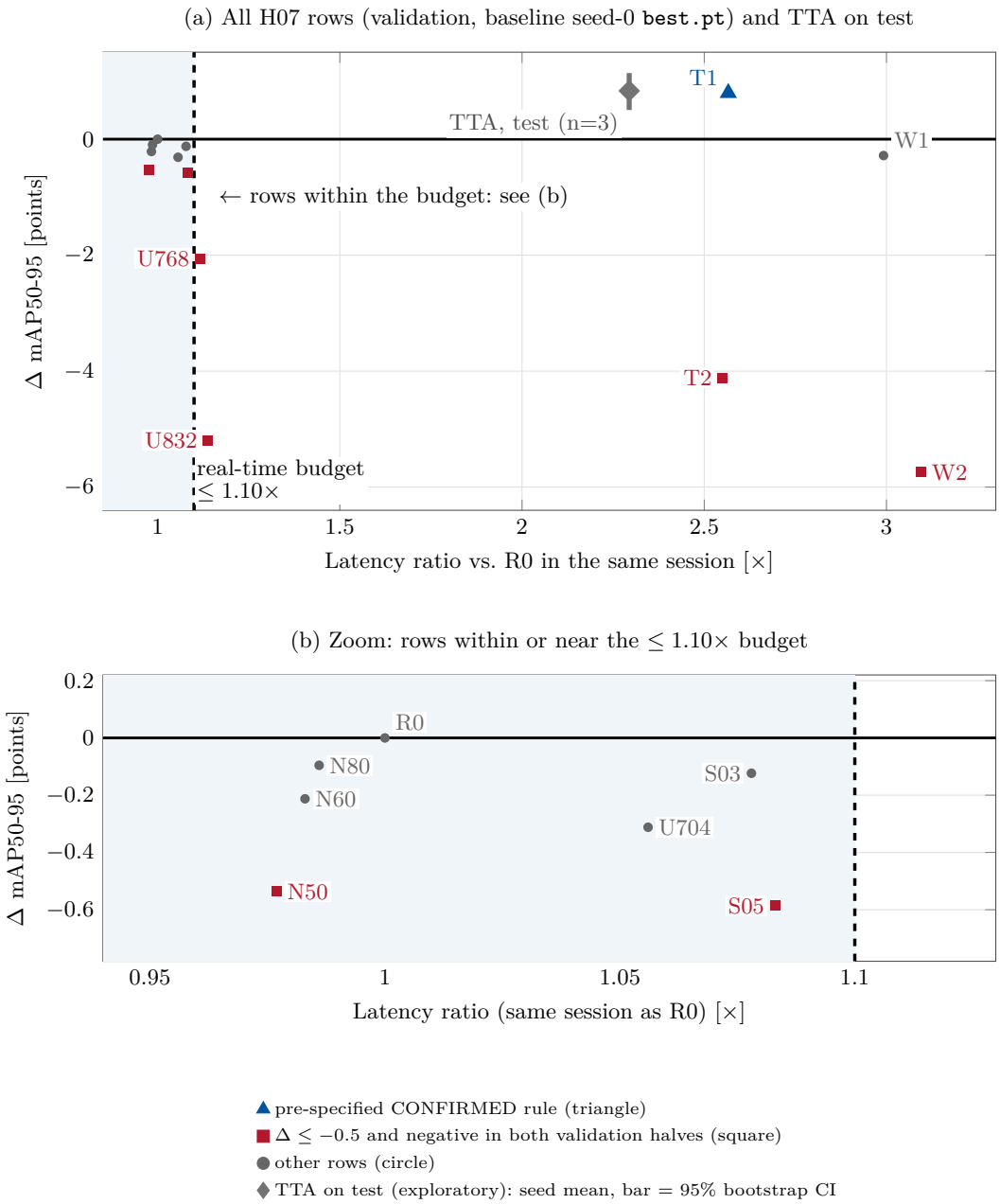


Figure 3 Accuracy–latency trade-off for the H07 rows (R0 reference, N: NMS-IoU, S: Soft-NMS, U: test resolution, T: TTA, W: WBF). Panel (b) zooms in on the real-time band. Triangle: confirmed; square: worse; circle: other. The thresholds are given in the main text.

Three explanations fit the same observation: a need for longer training; a mismatch of α, γ and λ_{cls}; or a pretraining–loss mismatch between the COCO-pretrained BCE head and the VFL targets. The two steps below partly separate these options.

Component decomposition: H05-pos and H05-neg.

Two differences separate VFL from the baseline BCE: negatives are suppressed by $\alpha\sigma(z)^{\gamma}$, and positives are weighted by t (Eq. (5)). We tested the two separately on VOC with two cards pre-defined

for S2 in the KITTI rule document. The setup was the same (YOLO11n, three seeds, validation only); the VOC test set was not opened. H05-pos ($\alpha = 1$, $\gamma = 0$) weights only the positives by t. H05-neg leaves the positives unweighted and only suppresses the negatives. A contribution is tested with a one-sided t test ($p < 0.10$). Additivity is tested with $I_s = \Delta_{\mathrm{H05},s} - (\Delta_{\mathrm{pos},s} + \Delta_{\mathrm{neg},s})$ and the pre-specified criterion $|\bar{I}| \leq 2\,\mathrm{SE}(\bar{I})$. H05-neg gave -1.81 points (-1.38, -2.20, -1.86; one-sided $p = 0.009$). H05-pos gave -0.52 (-0.26, -1.32, $+0.01$; $p = 0.16$) and did not meet the contribution criterion. $\bar{I} = -0.53$ (SE 0.46) meets the additivity criterion ($|\bar{I}| = 0.53 \leq 2\,\mathrm{SE} = 0.92$). At $n = 3$, however, this test cannot reject an interaction; it does not show that there is none. The loss comes mainly from negative suppression (descriptive share ≈0.63; seed range 0.48–0.73). The share of the positive component (≈0.18; −0.01–0.41) could not be separated from zero, and the unexplained share is ≈0.19. One possible explanation is a hypothesis, because the gradient norm was not measured per component. Early in training, most negatives have low scores. The factor $\sigma(z)^\gamma$ then weakens this term, which is thought to carry most of the gradient norm. The explanation applies to untuned VFL. It does not generalize to "negative suppression is harmful".

Gradient-magnitude compensation: H05c (exploratory).

H05c differs from H05 only in the classification loss gain: $\lambda_{\mathrm{cls}} = 0.5 \to 1.47$. This value brings the measured VFL/BCE gradient-norm ratio to 1 on average. The gain scales both terms together, and the ratio is not constant during training. The question is whether the loss comes from the magnitude or the form of the gradient. The decision rule was written before the runs, but after the KITTI results had been seen, in a separate document (2026-10-22; Section 9). The card makes no improvement claim. The rule tests $D_s = \Delta_{\mathrm{c},s} - \Delta_{\mathrm{H05},s}$ with a one-sided t test ($\bar{D} > 0$, $\alpha = 0.10$). The recovery is $R = (\bar{\Delta}_{\mathrm{c}} - \bar{\Delta}_{\mathrm{H05}})/(-\bar{\Delta}_{\mathrm{H05}})$. If $\bar{D} > 0$ and $p < 0.10$, then $R \geq 0.75$ means "magnitude explains the loss" and $0.25 \leq R < 0.75$ means "partly explains"; otherwise it does not explain it. The pre-specified expectation was "partly" ($R \approx 0.3$–0.7). Compensation reduced the loss to $\bar{\Delta}_{\mathrm{c}} = -0.51 \pm 0.59$ points ($\bar{D} = +2.36 \pm 0.47$, one-sided $p = 0.0065$, $R = 0.82$). The decision is "magnitude explains the loss", which is stronger than the expected "partly". The seed-level 95% t interval for R is 0.37–1.28, and the one-sided 90% lower bound is 0.63. Because the decision rests on a point estimate, the line between "magnitude" and "partly" is not sharp; seed 1 lies in the "partly" region. That at least half of the loss was recovered is supported at the pre-specified α level. The manipulation check passed: the realized effective gradient ratio is 1.02–1.03 in the three seeds.

Limits of interpretation.

The remaining difference covers zero at the seed level. We therefore cannot tell whether the remaining share comes from the form of the loss; we cannot claim that "form plays no role". The answer this card can give is exploratory. The VOC loss of untuned VFL is largely consistent with the classification gradient falling to ≈one third of BCE while λ_{cls} was not adjusted. This agrees with the H05-neg result and, among the possible causes, supports the mismatch of α, γ and λ_{cls}. Three things could not be separated: whether the recovery is specific to VFL (the gain also weights classification more heavily against the box and DFL terms), longer training, and the pretraining–loss mismatch.

7 KITTI generalization experiment

7.1 Setup

Questions and rules.

To test whether the VOC finding is specific to this dataset and model, we answered two pre-specified questions with the same cards. The results also show how sensitive an acceptance based on small Δ values can be to checkpoint selection. This second reading does not replace the answers to the questions. S1 (external validity): do the same seven training cards produce a within-setting Δ over the baseline in a different domain (driving scenes) and with two models? S2 (mechanism): does H05's loss on VOC come from suppressing negatives or from weighting positives with the IoU-aligned target? The rule document was frozen before the full queue started (SHA256 in the experiment log). Later changes were made before any KITTI card result was seen (Section 9). The claim boundary

is fixed in the document. Only within-setting Δ values are reported. There is no comparison of absolute mAP across datasets or models and no causal generalization claim.

Data, models and training.

KITTI [32] contains 7 481 labelled images and eight classes. We did not use the ready-made Ultralytics split, because it leaks: 106 of the 141 driving recordings appear in both training and validation. Our split unit is the *scene group*. A scene group merges recordings that share an image pair with normalized cross-correlation (NCC) ≥ 0.95 (141 recordings $\rightarrow$ 59 groups). Training, validation and test have 36, 10 and 13 groups (4 990, 987 and 1 504 images). The test split was locked and opened once, after all training had finished. `DontCare` regions are ignored in training. In evaluation, they ignore only detections that do not match a real object. This is not the official KITTI AP (Online Resource 1, Section C.1). We used two COCO-pretrained models. YOLO11s is from the same family at a larger scale. YOLO26n [9] has NMS-free end-to-end inference, an L1 box loss without DFL and two training heads. Version, epochs, resolution, batch and augmentation are the same as on VOC. With KITTI's few iterations, the `auto` optimizer would switch to AdamW, so we fixed the behaviour actually used on VOC (MuSGD, learning rate 0.01). Card definitions and copy-paste parameters did not change ($8 \times 3 = 24$ runs per model). All 48 runs passed the automatic health gate and the learning check. The validation values of the card runs were not read until the YOLO11s queue had finished. No published anchor exists, so the KITTI baseline did not pass a plausibility gate. We report the absolute test mAP50-95 only to give the scale of the Δ values (`DontCare` rule, mean of three seeds). It is 19.30 (`best.pt`) and 19.66 (`last.pt`) points for YOLO11s and 18.85 and 17.35 for YOLO26n. These values are not anchors and are not compared with VOC or across models (substitute checks: Online Resource 1, Section C.1).

Checkpoint and decision rules.

The primary endpoint is `best.pt`. It is selected by Ultralytics' validation fitness without `DontCare` and is the same for all cards. The `last.pt` weights of each run, i.e., the epoch-100 weights, were evaluated in the same single test opening. This is a fixed-epoch evaluation without validation-based selection. It is secondary and descriptive and does not enter the Holm families. If the $\bar{\Delta}$ of `best.pt` and `last.pt` has the same sign, the card is labelled "selection-independent"; otherwise "sensitive to `best.pt` selection". The label is not causal. The primary metric is the seed-paired Δ of the `DontCare`-rule mAP50-95 on test. Each model is evaluated separately, and there is no validation-based screening. The improvement family (H01, H03, H04, H06, H08) is tested with Holm-5 per model. Acceptance requires $\Delta \geq +0.40$ points and a Holm-adjusted $p < 0.10$. In the directional replication family (H02, H05; Holm-2), "replicated" means $\bar{\Delta} < 0$ and a Holm-adjusted $p < 0.10$. The p values entering Holm come from one-sided paired t tests. They are not defined like the two-sided p values in Table 4, and we do not call the KITTI acceptance criterion "significant". The decision rule does not include the sampling variance of the test scene groups. We therefore also report a clustered bootstrap interval over the 13 test scene groups ($B = 2000$). H07 was tested on validation only, and per model only the row with the highest validation mean entered the test. The H07 and `last.pt` results are exploratory.

7.2 Results

Answer to S1.

Table 5 gives, per card, the seed-paired Δ, the clustered CI, the Holm-adjusted p and the decision. In the improvement family, no card on YOLO26n met the acceptance criterion. On YOLO11s, only multi-scale training (H01) did: $+1.49 \pm 0.34$ points, Holm-adjusted $p = 0.042$. For the other four improvement cards of YOLO11s, the Holm-adjusted p does not fall below 0.252. The largest Δ observed on YOLO26n (H03) is below the card's own critical effect (1.83). For the cards that were not accepted, the upper bound of the seed-level t CI is at most +5.68 points on YOLO26n and +7.96 on YOLO11s. The design rules out only improvements larger than these. In the directional replication family, no card replicated. On YOLO11s, the Δ of H02 and H05 is positive, the opposite sign to VOC. The mechanism was active in every run (head counter), so the result is not evidence that "Varifocal is harmless". Under the pre-specified rules, the answer to S1 is as follows. In the improvement family, 0/5 cards on YOLO26n and 1/5 on YOLO11s

Table 5 KITTI test (13 scene groups, mAP50-95 with the DontCare rule): card − baseline Δ [points], $n = 3$ seed-paired. Blue+bold: card that meets the pre-specified acceptance criterion.

Card	Δ test ± s.d.	Clustered CI	p_{Holm}	Decision	Δ `last`	Robustness
YOLO26n (critical effect ≈ 3.61 points)						
H01 multi-scale	+0.12 ± 2.23	[−0.88, +1.49]	1.000	criterion not met	+0.32	independent
H02 Inner-CIoU	−0.93 ± 2.00	[−1.56, +0.18]	0.507	not replicated	+0.06	sensitive
H03 box copy-paste	+1.01 ± 0.66	[−0.12, +2.92]	0.289	criterion not met	+1.28	independent
H04 mosaic scale	−1.24 ± 2.18	[−2.02, +1.04]	1.000	criterion not met	+0.35	sensitive
H05 Varifocal	+0.61 ± 1.52	[−0.84, +1.41]	0.721	not replicated	+1.26	independent
H06 box loss gain	−0.25 ± 0.68	[−1.38, +0.71]	1.000	criterion not met	+0.10	sensitive
H08 high dose	+0.71 ± 0.70	[−0.45, +2.20]	0.444	criterion not met	+1.14	independent
YOLO11s (critical effect ≈ 2.92 points)						
H01 multi-scale	**+1.49 ± 0.34**	[−0.34, +2.28]	**0.042**	**ACCEPTED**	−0.66	sensitive
H02 Inner-CIoU	+1.14 ± 0.52	[−0.53, +1.99]	1.000	not replicated	+0.00	independent
H03 box copy-paste	+1.41 ± 1.07	[+0.35, +2.58]	0.252	criterion not met	−0.12	sensitive
H04 mosaic scale	+1.07 ± 0.72	[+0.18, +1.77]	0.252	criterion not met	−1.21	sensitive
H05 Varifocal	+2.28 ± 0.76	[+0.52, +3.61]	1.000	not replicated	+0.03	independent
H06 box loss gain	+0.57 ± 0.56	[+0.10, +1.83]	0.252	criterion not met	−0.47	sensitive
H08 high dose	+1.68 ± 2.53	[+0.89, +2.76]	0.252	criterion not met	+0.51	independent

Acceptance: $\Delta \geq +0.40$ and $p_{\text{Holm}} < 0.10$ (Holm-5; directional Holm-2 for H02 and H05; per model). Clustered CI: bootstrap over the 13 groups ($B = 2000$). It does not include seed variance. Seed t-CI $= \bar{\Delta} \pm 2.48\, s$. Δ last is descriptive. Robustness: independent if best and last have the same sign.

met the acceptance criterion. The VOC losses of H02 and H05 did not recur on either model. Under the claim boundary of the rule document, this is neither a "generalizes" nor a "does not generalize" result. The robustness of the single acceptance is examined below.

Answer to S2.

S2 was answered with the two component cards on VOC, using their pre-specified criteria (Section 6). H05's VOC loss comes mainly from suppressing negatives. Weighting the positives by t did not meet the contribution criterion, and the two components met the additivity criterion. This answer holds for VOC and YOLO11n. On KITTI, the same intervention caused no loss. On both models it peaked later than the baseline (6/6 seeds; Online Resource 1, Section C.3). An untested hypothesis on KITTI is that the classification gradient weakened by VFL slows down fitting. This could cause a loss for the VOC baseline, which is still improving at 100 epochs. For the early-peaking KITTI baseline, it could produce a different effect under `best.pt` selection.

Fragility of the single acceptance.

The YOLO11s H01 acceptance is reported under the pre-specified rule, but four findings make it fragile. Items (i) and (iv), the clustered bootstrap interval and the three-class subset are pre-specified secondary analyses. The hierarchical bootstrap and the scene-group and tram sensitivity analyses are post-unlock and exploratory. None required a new test opening or new training (Online Resource 1, Section C.5). (i) The `last.pt` Δ is −0.66. The sign flips, and the card is labelled "sensitive to `best.pt` selection". (ii) The clustered bootstrap interval over the 13 test scene groups, [−0.34, +2.28], crosses zero. A hierarchical bootstrap that resamples seeds and scene groups together gives [−0.67, +2.61] and $P(\Delta \leq 0) = 0.077$. (iii) Leaving out the drive_0056 scene group lowers Δ to +0.45. All 126 tram objects in the test split are in this group. Leaving out the tram class gives $\Delta = +0.53$, and the gain does not appear in the three standard classes (car, pedestrian, cyclist). Leaving out the same group gives a similar change on YOLO11s for H02 and H05 (+1.02, +0.97; H01 +1.04). This fits a weakness of the baseline in this group better than a card-specific effect. (iv) On validation, the acceptance criterion on YOLO11s was met by H04, not H01. These analyses do not change the

pre-specified decision. They show that the acceptance is sensitive to checkpoint selection and to a single scene group.

Training-free inference (H07).

On YOLO26n, the U832 row selected on validation (832 px test input) gave +1.54 ± 0.58 points on test. With a latency ratio of 1.008, it stayed within the ×1.10 budget. On YOLO11s, the selected W2 row (WBF fusion of the T2 branches) gave +2.02 ± 0.95. It raised latency to ×3.91, however, and fell outside the budget. Both are single rows and exploratory (no CI or Holm; untested hypothesis for the contrast with VOC: Online Resource 1, Section C.6).

8 Noise sources behind small differences

This section contains no new analysis. It groups the findings under the four noise sources of the central thesis and derives a recommendation for each.

(i) Seed variance and resolving power.

The validation s.d. of the baseline is of the same magnitude as the effects expected in advance (Section 1). With three seeds, the pooled critical effect is ≈1.2 points, and 1.58 points at 80% power (Section 3.3). On KITTI, the card mean of the critical effect at 50% power is 3.61 points on YOLO26n and 2.92 on YOLO11s. For YOLO11s H01 it is 0.95, owing to a low seed s.d. The effects expected in advance (Table 1) are below these thresholds. For five of the seven VOC cards, the 95% interval covers both the screening and the acceptance margin (Section 5.1). The results 0/7 and 1/10 are therefore not evidence of no effect. A single outlier seed can also shift the result (seed 1 in H04; [8]). *Recommendation:* set the number of seeds in advance with a resolving-power calculation, and report the critical effect together with the result.

(ii) Validation sampling.

On the 1000-image validation set, the image-sampling error of the per-seed Δ is ≈0.50 points. This is above the +0.30 screening threshold, so the seed-0 gate cannot resolve +0.30. With zero effect, a card passes with an estimated probability of 10.5–27.5%. The validation s.d. of the baseline is 3.67 times that on test (F test, one-sided $p = 0.069$). At $n = 3$ this difference cannot be resolved; the image-sampling error is more informative. H03's increase at seed 0 and its sign change on test are consistent with this noise. A card that enters the test because it has the highest validation mean may look optimistic owing to regression to the mean (the winner's curse) [7, 8, 22]. The disappearance of H03's mAP50 gain on test, however, is not fully explained by this noise model. *Recommendation:* make the validation set large enough that the per-seed sampling error falls below the effect of interest, and re-measure the card selected on validation on a locked test set.

(iii) Single-seed screening.

The apparent accuracy of the screening gate is not a finding. That none of the six screened-out cards received KEEP with three seeds (6/6) is expected with probability ≈0.77 even with zero effect (Monte Carlo; Appendix A.4). An uninformative gate would screen out all six zero-effect cards with probability 0.145–0.512. The gate saw no true positive, so its sensitivity cannot be measured. The gate also carries a false-negative risk. In the independent seed 1–2 mean, no screened-out card exceeds +0.30. The closest is H06, which was screened out at seed 0 (+0.27; −0.42 at seed 0). The short pilot did not predict the final ranking either (H04; a single-card example). *Recommendation:* treat single-seed screening only as a candidate generator and the short pilot only as a code-health gate, and decide with at least three seeds.

(iv) Checkpoint selection.

On KITTI, we define the common shift of the cards as the mean test mAP50-95 of the seven cards minus the baseline mean. On YOLO11s it is +1.38 points with `best.pt` and −0.27 with `last.pt`. On YOLO26n it is +0.01 and +0.65. Switching the checkpoint thus moves this shift by ≈4.1 times the acceptance threshold on YOLO11s and ≈1.6 times on YOLO26n. The robustness label came out as "sensitive to `best.pt` selection" for 7 of the 14 cards (Table 5). On YOLO11s, the `best.pt` Δ is positive for all seven cards, including H02 and H05, which were negative on VOC. The `last.pt` Δ values scatter around zero. On YOLO26n the pattern is mirrored (Online Resource 1, Fig. C1). The seven cards were measured against a single

baseline, so 7/7 is not seven independent pieces of evidence. That the clustered CI excludes zero for five cards on YOLO11s also fits this common shift better than a card effect. A remaining explanation is a hypothesis. The validation set is small, with only 10 scene groups. On YOLO11s, `best.pt` selection often picks the epoch-1 weights. It may thus catch a peak that is specific to this set and only partly carries over to test (winner's curse and scene-group noise). Other explanations and a diagnosis of the training pipeline are in Online Resource 1, Section C.4. The experiment that would separate them (fixed-epoch evaluation at intermediate epochs, or more seeds) was not run. On VOC, the baseline's `best.pt` lies near the end of training (best epochs 93, 100 and 84). The `last.pt` sensitivity of the cards was not measured on VOC. *Recommendation:* report `best.pt` together with the fixed-epoch weights (`last.pt`).

Possible explanations for the defaults (hypotheses).

No experiment was run to separate them, so we offer three explanations, at the level of hypotheses, for why no training-time intervention was accepted. The first is the assignment structure. TAL scales the class target of the positives by the prediction–GT IoU [17], so the baseline BCE loss already works with an IoU-aware target. Inner-CIoU and the box loss gain may have upset an established box–class balance. That AP_{75} did not move in either case is consistent with this. A related explanation for H05 was tested in an exploratory way: gain compensation recovered ≈82% of the loss (a single compensation value, VOC validation). The second is the scale axis. mAP50-95 did not rise for either of two neighbouring interventions. The default amplitude may suit this regime, although we cannot say it is "optimal". The recommendation to weaken augmentation for small models [6] was not reproduced here. The third is the regime. In a 100-epoch fine-tuning, COCO pretraining covers the 20 VOC classes [24]. The gap that extra augmentation diversity could fill may therefore have been narrow. Writing the hypotheses, thresholds and test plan before the runs [23] kept the negative results and the deviations visible.

9 Limitations

This study evaluates augmentation, loss and inference-time interventions in the fine-tuning regime within a protocol fixed in advance. It nevertheless has some methodological and experimental limitations.

Statistical power and sampling.

The three-seed ($n = 3$) design is a practical compromise within the target compute budget. On the VOC validation set (1000 images), however, it resolves with high power only clear effects of $\gtrsim 1.0$ point. Cards that did not pass the acceptance threshold should therefore not be read as having no effect. They should be read as showing no statistical advantage at the $\Delta \geq +0.30$ threshold under the observed variance. The sampling variance of the 13 scene groups in the KITTI experiment is not part of the primary decision rule; it is addressed by a separate clustered analysis. In addition, the absolute VOC values are computed with the 101-point COCO metric and exclude `difficult` objects. They cannot be compared directly with the original 11-point VOC devkit numbers [28].

Experimental scope and data regime.

The findings are limited to small COCO-pretrained YOLO architectures for real-time detection (YOLO11n, YOLO11s, YOLO26n), a 640 px input resolution and a single-GPU setup. The model family and the dataset changed together, so their separate contributions were not isolated. On datasets without segmentation masks, we applied box-based pasting. This departs from the conditions of the pixel-level masked augmentation proposed by Ghiasi et al. [5].

Protocol discipline and transparency of records.

The pre-planned hypothesis cards are kept clearly apart from the diagnostic analyses run after the test was opened (extra seeds, the WBF grid search and the H05c analysis). Exploratory findings are not counted as primary evidence. All run steps, the behaviour of the stopping rule and the protocol records are documented in full in Appendix A and Online Resource 1.

10 Conclusion

Small-gain claims could not be separated from noise under a typical three-seed fine-tuning budget. The paper shows this with seven interventions and two datasets and documents four sources of noise: seed variance, validation sampling, single-seed screening and checkpoint selection. The results apply to untuned single-point interventions against an untuned default recipe. In the PASCAL VOC 07+12 fine-tuning of COCO-pretrained YOLO11n, none of these interventions passed the pre-specified acceptance threshold. Copy-paste (H03), which entered the test, did not beat the baseline on test either. After the exploratory Holm-7 correction, the only intervention that stood out was untuned Varifocal loss, in the negative direction. Most of its drop was recovered by compensating the classification loss gain (exploratory). The single acceptance on KITTI was sensitive to checkpoint selection. The three-seed design cannot resolve smaller effects, so these results are not evidence of no effect.

The recommendations for the four noise sources are in Section 8. The findings are limited to two datasets, three small YOLO models, 640 px and a single GPU (Section 9). Future work could test the screened-out interventions with more seeds, a larger validation set and a tuned baseline. The `best.pt`–`last.pt` divergence could be examined with fixed-epoch evaluation at intermediate epochs. The Varifocal loss could be studied with sweeps over α, γ and the gain, and the VOC–KITTI difference with a YOLO11n×KITTI bridge experiment.

Declarations.

- **Funding:** This work received no funding from any institution or funding body.
- **Competing interests:** The authors have no competing interests to declare.
- **Ethics approval and consent to participate:** Not applicable. The study uses only publicly available image datasets (PASCAL VOC, KITTI) and involves no human or animal participants. KITTI was used under the CC BY-NC-SA 3.0 licence for academic purposes only, and no images were redistributed.
- **Consent for publication:** Not applicable.
- **Data availability:** The PASCAL VOC 2007 and 2012 and KITTI 2D object detection datasets used for training and evaluation are publicly available. The split lists, `DontCare` ignore regions and seed-level results derived in this study are in the code repository.
- **Code availability:** The code, hypothesis cards, decision-rule documents, run configurations, seed-level results, split lists, analysis scripts and Online Resource 1 are available at https://github.com/ahmetgozcelik/voc-yolo-augloss-analiz. The version reported in the paper is archived on Zenodo: https://doi.org/10.5281/zenodo.23797459. The experiments used Ultralytics v8.4.165 (`004db34`).
- **Author contributions (CRediT):** **A. G. Özçelik:** conceptualization, methodology, software, investigation, formal analysis, data curation, visualization, writing – original draft. **U. Asıl:** conceptualization, supervision, validation, writing – review and editing. Both authors read and approved the final manuscript.
- **Use of AI tools:** The authors used AI-based tools at various stages of this work (e.g., code development, analysis and language editing). All outputs were reviewed and verified by the authors, who take full responsibility for the content of the article.

Appendix A Experiment log summary and computation notes

This appendix gives the records and computations that a referee needs to check. It covers a summary of the run log, the application of the stopping rule, post-unlock work and the GPU budget. It also covers the computation of the screening probabilities, the gradient-norm ratio and the sources of the plausibility anchor. The run-log table, additional results and method details are in Online Resource 1 (also in the GitHub repository).

Table A1 in Online Resource 1 breaks down the runs per card. It covers the baseline plausibility check, the three rounds, the locked-test stage and the post-test additional-seed runs. Of the 60 run stamps in total, five did not end with exit code 0. Three are calibration runs (rc=98), one is an orphan process (rc=96) and one is an H05 failure (rc=95). There are 35 runs of evidential value. The first 22 are 12 stage-3 trainings, 1 H07 evaluation and 9 locked-test evaluations. The other 13 are 12

trainings for additional seeds and 1 WBF diagnostic run. The remaining 25 runs are gate, control, calibration and failure records and were not reported as results. A total of 75.45 GPU hours was spent (target 75, ceiling 90). The full command and rule parameters of each run are stored in the `config.json` file in the run directory. The queues are stored under `kuyruk/`.

A.1 Application of the stopping rule

Fig. A1a in Online Resource 1 shows the GPU time per round and the cumulative total. Panel (b) shows the Δ of the top training-time card in each round. Filled circles are three-seed means, and open circles are seed-0 screening values. The stopping rule says: move to the locked test if the top training card has $\Delta < +0.30$ in two consecutive rounds. It was triggered at the end of the second round, based on the values available at that date. The top training card of the first round was H03, and its three-seed mean (+0.25) was already known. The second round had only seed-0 screening runs, and its top card was H04 (−0.30). The cumulative time at that point was 35.30 hours. The second- and third-round means in the figure were computed with post-unlock additional seeds. They did not enter the stopping decision.

The rule text does not specify whether Δ is the seed-0 value or the three-seed mean. For each round, we used the best estimate available at the time. This was the three-seed mean in the first round (+0.25) and the seed-0 value in the second (−0.30). Had both rounds been read with the seed-0 value, the first-round value would have been the seed-0 difference of H03. That value is +0.37, i.e., $\geq +0.30$, so the rule would not have been triggered at this point. The rule text also does not distinguish training-time cards from training-free ones. In practice, the training-free TTA row of the second round was kept outside the rule. This row was on validation with seed 0, gave +0.80, and fell outside the $\times 1.10$ relative latency constraint. Had TTA also counted as a round-2 card, the rule would not have been triggered ($+0.80 \geq +0.30$). The stopping decision therefore rests on the training-time interpretation.

H08 was the only card planned for the dose question. It was run in the third round, after the rule had been triggered. Its screening value was +0.22 (3.16 hours), and its three-seed mean was −0.08. The study then moved to the locked test (0.21 hours). The design, code and run expectation of H08 had been written about 9.6 hours before the trigger, before any second-round result had arrived. The execution sequence was established from the queue and code file logs. The rule document set a limit of five rounds and contained a candidate pool for the third round. However, it had no pre-specified exception that allowed another round after the rule had been triggered. Running H08 is therefore a deviation from the pre-specified rules (Section 9). The card was not accepted, and the deviation did not turn any decision positive.

A.2 Post-unlock work

The following is the complete list of work done on VOC after unlocking. (i) Seed 1 and seed 2 validation runs of the six cards eliminated at screening (H01, H02, H04, H05, H06, H08), i.e., 12 training runs. (ii) An exploratory diagnostic grid for the WBF branch of H07 on the validation set. It varied the WBF threshold, the skip threshold and the fusion type. None of its settings was applied to the test, and the best point of the grid does not count as evidence. (iii) Analyses of n=3, power and chance. These used existing outputs and required no new runs. (iv) The two component cards of H05 (H05-pos, H05-neg), run under a second rule document (Section 6). They were run with three seeds on validation only (6 training runs), and the VOC test was not opened. (v) H05c, the λ_{cls} gain compensation card for H05, run under a third rule document (2026-10-22). It was run with three seeds on validation only (3 training runs), and the VOC test was not opened.

This list covers work on VOC data. The KITTI experiment was carried out on a separate dataset under a separate rule document (Section 7). After the KITTI test was opened, no new training, test inference or hyperparameter search was done. The pre-specified parts were the test evaluation of the 48 runs, the H07 test rows and the primary analysis. Everything else is exploratory or secondary. This includes the pre-specified secondary analysis on validation and the re-evaluation of `best.pt` and `last.pt` (96 evaluations). It also includes several analyses of the existing test predictions: clustered bootstrap p, Holm-10, Holm-14 and sign-flip sensitivity analyses, Δ without `DontCare`, class subsets

and secondary tables. The remaining items are the hierarchical bootstrap, Δ per scene group, the tram sensitivity analysis, the external reference value and a diagnosis of the baseline on 1000 training images (Online Resource 1, Section C). Apart from these, no other runs, hypotheses or hyperparameter searches were carried out on VOC, and the test was not touched. The additional seeds were added post-unlock, following the principle of not reporting numbers with n<3. They were given only to the six cards eliminated by the pre-unlock screening decisions.

A.3 GPU budget

After unlocking, the additional-seed runs of the six cards and the WBF diagnostic grid took 36.79 hours. They raised the total from 38.67 to 75.45 hours; totals were computed from unrounded values. This is slightly above the target (75) and below the ceiling (90).

The generalization experiment (Section 7) is not included in Table A1 (Online Resource 1) and was logged separately. It comprises 54 full runs, all with 100 epochs. On KITTI, two models $\times$ eight arms $\times$ three seeds give 48 runs. On VOC, H05-pos and H05-neg $\times$ three seeds give 6 runs. The total training time is 89.02 hours: YOLO11s 41.15, YOLO26n 28.99 and the VOC mechanism cards 18.88. The three runs of H05c (VOC, validation; 10.25 GPU hours) were logged separately and are not included in this total. Smoke tests, short pilots, H07 and test evaluation are also excluded. This total is therefore defined differently from the 75.45-hour total of the first experiment, which counts all run stamps. The KITTI test split was opened on 2026-10-22 at 12:02:15. It was opened in a single opening with the `best.pt` and `last.pt` weights of the 48 runs. The VOC test was not opened in this experiment.

A.4 Computation of the screening probabilities

The two probabilities in Section 5.1 answer different questions and are descriptive only.

(A) The first is the probability that an uninformative seed-0 gate ($\Delta_{k,0} \geq +0.30$) eliminates all six cards when their true effect is zero. It was computed as $(1-p)^6$, where $p = 1 - \Phi(0.30/\sigma)$ is the probability that a single card passes the gate. A single common σ is used, with two candidate values. The first is an image-sampling estimate, obtained by scaling the test bootstrap standard error of H03 to the validation size (0.502; $p = 0.275$). The second is the s.d. of the validation seed differences of H03 (0.240; $p = 0.105$). These give 0.145 and 0.512, respectively.

(B) The second probability uses the definition "an eliminated card is not accepted with three seeds either". Its value, 0.774, was obtained by Monte Carlo with 400 000 draws per card. Each draw has three seeds with $\Delta_s \sim N(0, s^2_{\Delta,k})$, where $s_{\Delta,k}$ is the observed s.d. of the card's seed differences. The true effect is exactly zero. A card is accepted if its mean $\Delta \geq +0.40$ and the one-sided $t > 1.886$. Cards were drawn independently. Two dependencies were not modeled: the correlation arising from the common baseline, and the inclusion of the screening seed in the $n = 3$ mean.

Neither computation proves that the gate is or is not informative. The sensitivity of the gate cannot be measured, because no true positive was observed.

A.5 Gradient-norm ratio

The VFL/BCE ratio reported for H05 is computed from the L2 norm of the loss gradient with respect to the classification logits. For the same logits and targets, the VFL and BCE gradient norms are measured every 200 training steps, giving 486 measurements per run. The ratio of the norm sums is then taken. The effective gradient ratio in the manipulation check of H05c is this ratio multiplied by 1.47/0.5 (Section 6).

A.6 Sources of the plausibility anchor

Two YOLO11n sources give the lower end of the anchor in Section 4.1. ADL-YOLO reports training from scratch, whereas YOLO-Master does not specify the initialization [29, 30]. The YOLOv8n value at the upper end (63.5) was verified in the text of Section 4.4 of the source. The corresponding table cell could not be read. The source uses the VOC07 test set as its validation or test set, so its checkpoint was probably selected on the test set [31]. The value of 60.2 for YOLOv8n trained from scratch lies outside the anchor and is listed for context only (Table 3).

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